Frontier



Hourglass Climate

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Hourglass Climate

Organization location (we welcome applicants from anywhere in the world)

Experimental work for this project will take place at Hourglass Climate's MARINER lab in San Pedro, California. Hourglass Climate's headquarters are based in Montclair, New Jersey.

Name(s) of primary point(s) of contact for this application

Tricia Light

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Hourglass Climate is a 501(c)3 nonprofit research organization dedicated to critical, independent science for responsible mCDR development.

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Evaluating the impact of secondary precipitation on OAE efficiency

[1] We use "project" throughout this template, but the term is not intended to denote a single facility or experimental setup. The "project" being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Overview

Ocean Alkalinity Enhancement (OAE) has outsized potential for impact as a scalable and effectively permanent (>10,000 years) form of CDR. However, the formation of secondary minerals during addition can consume a significant amount of the added alkalinity, potentially lowering the CDR efficiency of an intervention. Here, we aim to conduct the most rigorous experiments on secondary precipitation to date to evaluate the matrix of field conditions under which precipitation can occur, as well as the long-term impact on CDR when precipitates form.

The problem

Lab-based experiments and field trials suggest that secondary precipitates such as brucite form under certain OAE conditions, but more work is needed to constrain the scenarios in which they precipitate and may subsequently dissolve. This latter information is vital because if secondary minerals formed during alkalinity addition ultimately dissolve, they may have minimal net effect on CDR efficiency and may not warrant mitigation measures to prevent.

Some secondary precipitates that form during alkalinity addition are assumed to quickly dissolve upon mixing with receiving waters, and they are thus considered to have little impact on CDR efficiency. Groups conducting OAE largely target conservative outfall conditions to avoid more extensive secondary precipitation, and CDR efficiency estimates generally assume that runaway secondary precipitation constitutes a durable alkalinity loss. While necessitated by the current state of the science, these approaches pose significant limitations for the scale-up and cost-efficiency of OAE. We must better understand secondary mineral responses under varied conditions to determine the circumstances in which secondary precipitation does and does not limit OAE efficiency and deployment at scale.

Investigating the ultimate fate of OAE secondary minerals using traditional scientific methods is challenging. Field trials limit the breadth of environmental conditions that can be explored at once, and monitoring is difficult and expensive. Meanwhile, traditional lab-based, closed system experiments often fail to adequately simulate the open system of real-world deployments. Secondary mineral responses likely vary between different locations and OAE approaches, so we must efficiently explore a broad solution space while targeting our research efforts toward work already underway.

Our solution

Hourglass Climate is uniquely positioned to evaluate the impact of secondary precipitates on OAE efficiency due to our deep technical expertise, analytical infrastructure, and one-of-a-kind experimental facilities. First, we will conduct laboratory experiments to constrain the field conditions under which secondary precipitates form. Next, we will evaluate the ultimate fate of these secondary precipitates in experiments that realistically simulate conditions in the water column and at the sediment-water interface.

We will conduct these dissolution experiments using a custom-built seawater-sediment column reactor system (Figure 1) in our **MARINER** (**Mar**ine **I**nnovative **N**egative **E**missions **R**esearch) Lab, housed in the Southern California Marine Institute (SCMI). This novel system enables us to achieve the high analytic resolution of lab-based experiments while realistically simulating open system OAE in the field, overcoming many shortcomings typically associated with lab-based experiments. We will use this system to conduct a series of incubation experiments, each of which will simulate field conditions in a region of interest to OAE project developers.



Figure 1. Hourglass Climate's custom-built seawater-sediment column reactor system.

Impact to the field

As Frontier has identified in its Target Innovation Areas, secondary precipitation during alkalinity addition is a key barrier to rapid scaling of OAE. Alkalinity consumed by the formation of secondary minerals is largely assumed to constitute a 1:1 loss in OAE efficiency. This assumption has far-reaching implications for OAE's feasibility as a climate solution at scale, both in terms of limiting alkalinity addition methods, rates, and project locations, and for CDR accounting and financing. By rigorously evaluating both precipitate formation and the assumption of durable alkalinity loss under varying field conditions, we

will determine if there are circumstances in which secondary precipitation does not actually reduce CDR potential, thereby broadening the potential for scaling OAE.

Our research will produce two key outputs for the benefit of the broader mCDR community:

1. We will generate a matrix detailing secondary precipitation thresholds under different conditions such as alkalinity feedstock, receiving water chemistry, and temperature. We will use publicly available information from OAE companies to guide initial starting conditions, and we will explore a wide range of plausible discharge scenarios to identify likely drivers of secondary precipitation in the field. Our findings will help project developers make informed decisions regarding optimal addition rates, siting, scalability, and more.

2. We will assess the ultimate fate of secondary minerals under realistic field conditions. If precipitates form during alkalinity discharge but dissolve in the water column or at the sediment-water interface, their net impact to OAE efficiency may be minimal. Short-term beaker experiments likely overstate the extent to which secondary precipitation poses a threat to rapid alkalinity addition rates. If we find that most secondary precipitates dissolve under a given range of environmental conditions or alkalinity addition methods, our work will accelerate competitively priced, scalable OAE in those contexts.

The work described above also gives us an opportunity to address another major uncertainty facing the OAE field: how alkalinity enhancement may impact naturally occurring alkalinity fluxes at the sediment-water interface. Both electrochemical and mineral-based OAE approaches at scale involve exposing sediments to higher alkalinity seawater. Biogeochemical processes in sediments may consume a significant amount of this added alkalinity, or OAE may reduce the natural alkalinity that sediments would have generated without intervention (i.e., “the additionality problem”, Bach et al. 2024). Roughly 20% of the alkalinity input to the ocean comes from the sediment column (Middleburg et al. 2020), and impacts to this natural flux will have major implications for the net impact of OAE. To most effectively and efficiently scale OAE, we must better constrain these uncertainties across different alkalinity additions and sediment types. As a cobenefit of the experimental design proposed here, we would be able to study these processes and potentially expand the range of sites and conditions under which OAE would be effectively implemented.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you’re differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

We aim to 1) identify alkalinity addition thresholds at which brucite and other secondary precipitates form in the water column and 2) use our custom-built reactor system to comprehensively assess the impact of secondary precipitation on CDR efficiency.

Phase A: Small-scale secondary precipitation experiments

To comprehensively assess long-term secondary mineral responses, we must first determine what secondary precipitates we expect to see under different OAE addition scenarios. We will conduct a series of small-scale bench experiments to determine how secondary mineral precipitation is driven by factors such as alkalinity feedstock, receiving water chemistry, and temperature. These experiments will produce a valuable standalone dataset and allow us to generate realistic secondary precipitates for Phase B.

We will use public information from two point-source OAE project developers, Ebb Carbon and Planetary Technologies, to constrain realistic alkalinity addition conditions. Ebb Carbon uses electrochemistry to produce NaOH from seawater. This NaOH is then mixed with seawater brine and returned back to the ocean (Savoie et al. 2025). Planetary uses an alkaline mineral feedstock consisting primarily of MgO (Planetary Operations and Monitoring Report 2026, Isometric). Ebb and Planetary have published data on their discharge parameters through peer-reviewed publications and on the Isometric carbon removal registry, respectively. Our experimental conditions will also be informed by the existing academic literature on secondary mineral precipitation (e.g., Hartmann et al. 2022, Hashim et al. 2025, Moras et al. 2022, Ringham et al. 2024, Shaw et al. 2025, Suitner et al. 2024).

We will conduct discrete short-term beaker experiments under various conditions (Table 1). For each experiment, we will add aqueous alkalinity simulating a realistic point-source OAE discharge and track the alkalinity thresholds at which secondary precipitates form. We will stir beakers to simulate seawater mixing, and we will explore alkalinity addition speeds and concentrations. Our experimental design will encompass current operating conditions at Ebb and Planetary as well as a range of plausible scenarios as similar OAE approaches scale up. We will vary standard oceanographic parameters (i.e., temperature and salinity) as well as other variables that may influence precipitate formation (e.g., total suspended solids, Ringham et al. 2024). These beaker experiments are generally intended to capture secondary mineral precipitation in the near-field mixing zone.[1]

[1] Here, we will assess secondary precipitates that form in minutes to hours, before atmospheric equilibration. Other precipitates (e.g., aragonite) may form more slowly, but seawater mixing minimizes their relevance in the field.

Table 1. Sample experimental parameters for beaker experiments to characterize secondary precipitates.

Parameter	Treatments
Alkalinity source	• NaOH • Supernatant from a seawater slurry with commercially sourced MgO
Salinity	• 25, 30, 35, 40 psu
Temperature	• 5, 10, 15, 20, 25, 30 ℃
Total suspended solids	• Low, medium, and high background particle loads

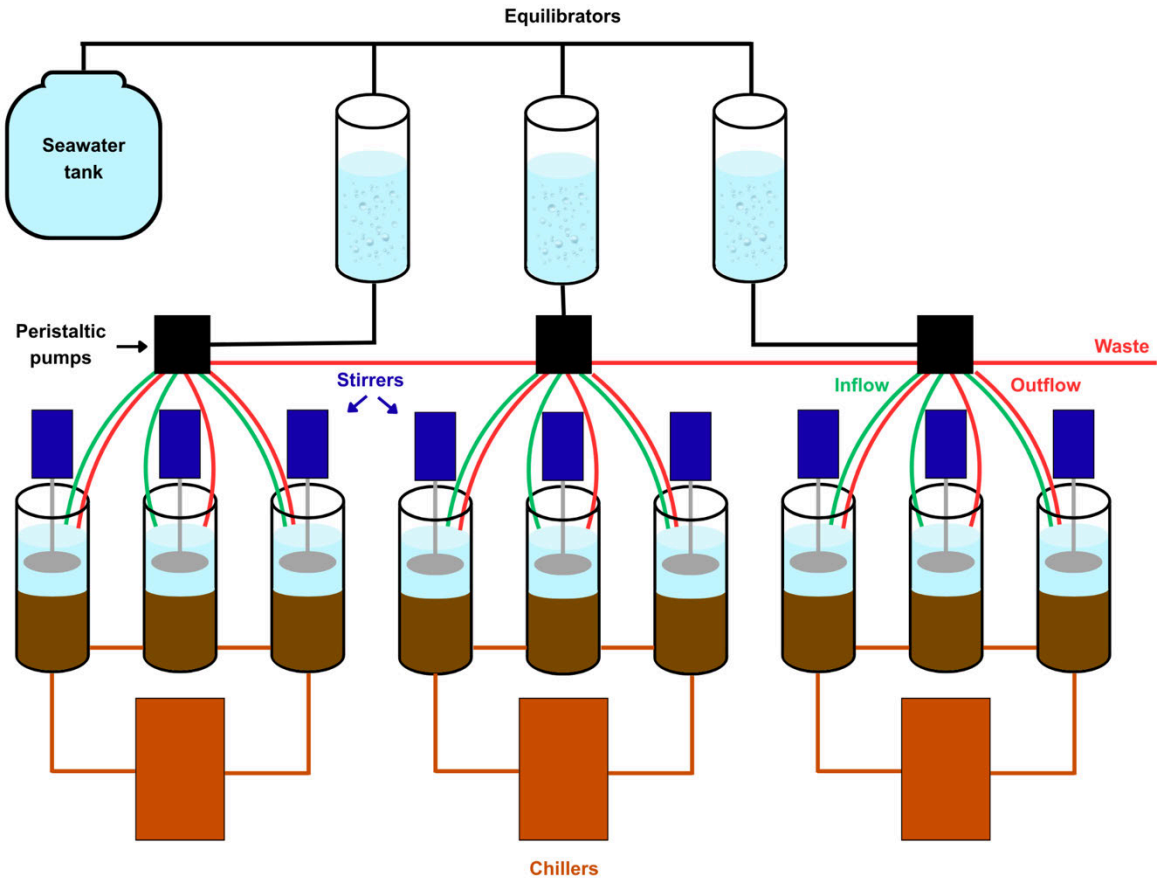
For each experiment, we will analyze turbidity, pH and total alkalinity to identify secondary precipitation thresholds. We will iteratively revise experimental parameters and conditions as needed. Select samples will be filtered for solid-phase precipitate characterization via scanning electron microscopy (SEM), elemental dispersive X-ray spectroscopy (EDS), and x-ray diffraction (XRD). Methods development and set-up will take 1 month, experiments 1 month, and sample and data analysis 2 months.

These experiments will help guide alkalinity addition rates and methods, and they will allow us to create secondary minerals for Phase B. Mineralogical composition, grain size, and morphology all influence dissolution rates (e.g., Shaw et al. 2025), so characterizing secondary precipitates under different OAE scenarios is vital to constraining real-world secondary mineral responses. We will identify representative end members for different addition scenarios, and we will determine how to consistently produce these precipitates in small batches.

Background: Hourglass Climate's seawater-sediment column reactor system

Hourglass Climate is uniquely positioned to investigate secondary mineral responses due to our custom-built OAE research infrastructure. We have developed a novel seawater-sediment column reactor system (Figure 2) to conduct experiments that realistically simulate field conditions while allowing for a high experimental control and analytic resolution.

Figure 2. High-level schematic of Hourglass Climate's seawater-sediment column reactor system (not to scale).



The system's 9 reactors can be filled with any target ratio of natural or artificial seawater, sediment, and OAE feedstock. Reactors can be run for any length of time, and a continuous supply of seawater facilitates long-term experiments without introducing closed system effects (e.g., evaporation, oversaturation of dissolution products, etc.). Seawater inflow rate can be adjusted to achieve water residence times from minutes to weeks. Incoming seawater is equilibrated with the atmosphere to stabilize carbonate chemistry, and continuous data loggers measure reactor temperature and atmospheric pCO_2. The sediment component of the reactors establishes more realistic geochemistry in the overlying water column, and reactors induce porewater advection to investigate under-appreciated alkalinity dynamics across the sediment-water interface. The reactor design allows measurement of overlying water and sediment porewater geochemical profiles at any temporal frequency.

See Section 2a for reactor system proof-of-concept data.

Phase B: Reactor experiments to assess secondary mineral responses

In Phase B, we will use our novel reactor system to evaluate the fate of secondary precipitates while realistically reproducing field conditions in the water column and at the sediment-water interface. Recent work suggests that dissolution of secondary phases can occur under certain conditions (Moras et al. 2024, Shaw et al. 2025, Varliero et al. 2024). We will conduct multiweek experiments that simulate alkalinity enhancement with and without secondary precipitates. We will not only constrain the short-term impact of alkalinity additions but also the longer-term fate of precipitates and recovery of the system after alkalinity addition. To determine how alkalinity dynamics vary between geochemically diverse potential OAE deployment sites, we will use at least two sediment types: silicate-rich sediments common in much of the USA and carbonate-rich sediments prevalent in equatorial regions where mCDR companies may scale up.

For each experiment, we will add equal volumes of natural coastal sediment and seawater to reactors. We will run the full system (i.e., stirring, seawater in and outflow) for at least 10 days to allow reactors to geochemically stabilize. Following stabilization, we will add alkalinity to the overlying seawater. Initial alkalinity additions will target realistic seawater concentrations ~3 meters from a point-source alkalinity discharge, and seawater inflow will dilute this addition over time (simulating seawater mixing). The reactor system allows for a high degree of flexibility and experimental control, so we will iteratively adjust variables such as alkalinity addition rate and water flow as needed. We will run the full reactor system for at least 3 more weeks to assess mineral responses and geochemical system recovery.

Reactors will receive alkalinity with and without precipitates to assess OAE efficiency impacts. To create the alkalinity additions, feedstocks will be added to a small batch of seawater, simulating alkalinity discharge and secondary precipitation at a point-source. This high-alkalinity water will be physically filtered to remove precipitates and added to aqueous alkalinity only experimental treatments. Other treatments will receive both high-alkalinity water and co-occurring precipitates, while others will only receive precipitates. We will then compare alkalinity concentrations and carbonate system impacts between aqueous and solid-phase alkalinity additions relative to controls with no addition.

We will monitor reactors through periodic sampling of seawater and sediment porewater. We will obtain in-house high-precision pH, total alkalinity (TA), and dissolved inorganic carbon (DIC) measurements, as well as trace metal, major cation (Mg, Ca), and nutrient data through external labs. After experiment take down, we will conduct sediment analyses such as SEM and XRD. Since the reactors are an open system and sediment-water interactions introduce additional complexity, leveraging these multiple lines of evidence to resolve secondary precipitate responses will be crucial. We have already developed a numerical framework that is representative of our reactor system, and we will leverage this framework to quantitatively constrain secondary mineral dissolution, carbonate system dynamics, and OAE efficiency.

Phase A will inform our exact experimental design. If we observe distinct secondary precipitates, we will assess responses for each precipitate end member (Figure 3). We will use our 9 reactors to evaluate precipitate dissolution under multiple scenarios; for example, we may assess different precipitate loads, alkalinity concentrations, and secondary precipitates incorporated into the sediment column by dynamic coastal processes. Our first

reactor experiment will evaluate precipitate responses in silicate-rich sediments, and our second experiment will use carbonate-rich sediments.

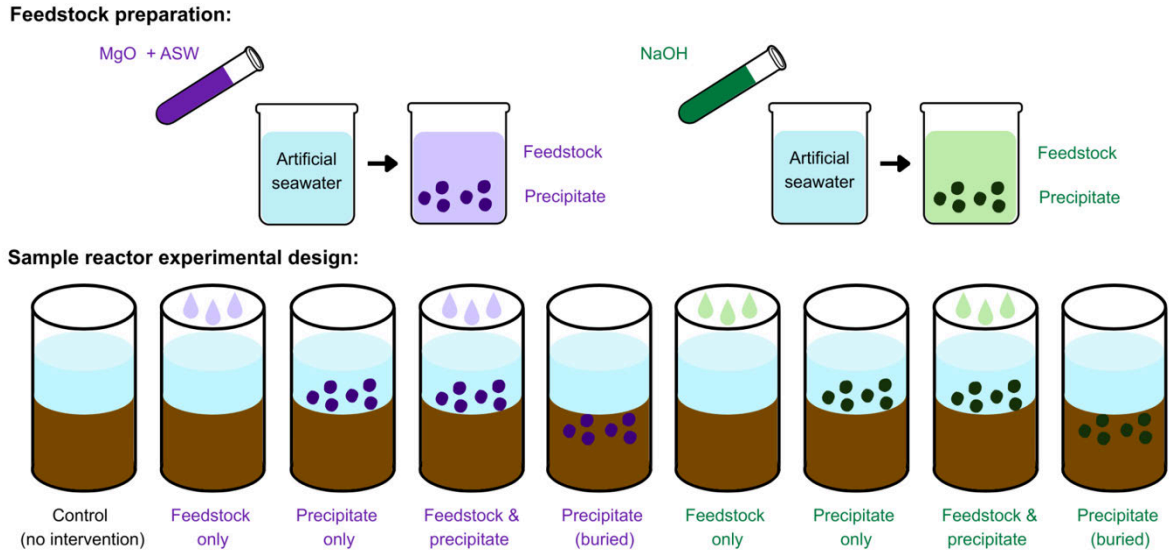


Figure 3. Sample experimental design for a seawater-sediment column reactor experiment.

These first-of-their-kind experiments will constrain secondary mineral responses while simulating the complex biogeochemistry of real-world OAE. By using realistic secondary precipitates, natural coastal sediments, and an open-system reactor design, we will generate uniquely valuable datasets for informing effective and efficient OAE deployments. These experiments will also produce critical information regarding how OAE impacts alkalinity fluxes at the sediment-water interface.

Each experiment will take 3 months to complete, accounting for setup and takedown. Experiments will be conducted in series, with 1 month for process improvements between experiments 1 and 2.

Data synthesis and dissemination

During the final 7 months of our project, we will analyze and synthesize our data. We will disseminate our findings through data repositories, conference presentations, and academic manuscript(s) submitted for peer-review.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Scientific

As with all scientific research, our work presents uncertainties. For Phase A, we must determine the best methods for assessing the sensitive boundary conditions between rapid mineral precipitation and dissolution. We will iteratively adjust parameters such as receiving water volume, stir rate, and alkalinity addition rate to overcome these risks and rigorously assess precipitation thresholds. We must also conduct method development on how to most effectively isolate and characterize secondary precipitates, and we may need to iterate on larger-batch mineral synthesis. We have accounted for this method development in our timeline and budget, and our team possesses the scientific expertise to address issues as they arise.

For Phase B, our reactor experiments pose the risk that lab-based work is never a perfect analog for the field. We are mitigating this risk by realistically simulating field conditions wherever possible: our custom-built reactor system facilitates water circulation, sediment water interactions, and more, and we will use realistic secondary precipitates. As such, our work will investigate secondary mineral responses under the most realistic conditions to date while still retaining the broad applicability and high analytic resolution of the lab. Another challenge during Phase B will be resolving a measurable precipitate dissolution signal above baseline processes. We will iteratively adjust our experimental design, water residence time, and sampling plan as needed to effectively address our research questions.

Technical

Technical infrastructure like our reactor system always poses a risk of mechanical or electrical issues. If issues arise, we will seek expert support from Blue Sonde, the engineering firm we contracted to design and install the system.

Project execution

We must determine appropriate locations, collection methods, permits, and transportation for Phase B reactor sediment. Sediment collection and timely transportation poses logistical and operational challenges, but our team has experience successfully navigating these processes. As a contingency, we have already established permissions and access for collecting natural sediments from the LA Harbor through SCMI.

MRV

We will not conduct MRV to generate carbon credits, but high-quality measurements are vital to our research plan. Small carbonate chemistry signals are challenging to resolve, but we will mitigate this risk through state-of-the-art sensors and high-precision bottle analyses, which we already reliably conduct in-house. We will leverage an existing numerical framework to translate these measurements into corresponding alkalinity enhancement and CDR efficiency.

Ecosystem

Our lab-based experimental plan poses no ecological risks.

Financial

Prices for equipment, consumables, and external analyses may rise over the timeline of this proposal. Most purchasing will be conducted early in our timeline to mitigate this risk, and we can adjust our experimental design to reduce the number of external analyses if necessary.

Unanticipated method development or system failure could impact timeline and thus salary cover. We mitigate these risks, as well as unexpected repair or replacement costs, through conservative budgeting, contingency allocation, and phased experimental design.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	Datasets for Phases A and B providing comprehensive guidance on methods to mitigate the impact of secondary minerals on OAE efficiency, at least 1 peer-reviewed publication synthesizing our findings.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	These deliverables will constrain circumstances in which secondary precipitation does and does not limit OAE efficiency and deployment at scale.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	This research will be complete once full experimental datasets are released and manuscript(s) are published on a pre-print repository and submitted for peer-review.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$497k